SHORT · DELTA LAKE VS. PARQUET

Delta Lake vs. Parquet

Parquet is a file format. It stores data by column, so files stay small and reads are fast. Delta Lake keeps those Parquet files and adds a transaction log called the delta log.

Key points

- A folder of Parquet files has no record of what changed.
- In Delta, every write becomes a new version in the _delta_log folder.
- That log gives you ACID transactions, schema enforcement, and time travel.
- It also enables MERGE, UPDATE, and DELETE. Plain Parquet has no history and no safe updates.

Tiny example

Save a small range of numbers as a table, which is Delta by default. Delete one row, then run DESCRIBE HISTORY to see both versions. In Databricks Free Edition, click New, then Notebook, and run each cell with Shift + Enter. No live output is recorded here.

```
df = spark.range(5)
df.write.mode("overwrite").saveAsTable("nums")   # Delta by default

spark.sql("DELETE FROM nums WHERE id = 0")
display(spark.sql("DESCRIBE HISTORY nums"))      # version 0 and 1
```

Misconception check

> Delta is not a different file format that replaces Parquet. The data is still Parquet files. The transaction log on top is what makes it a reliable table.

Remember

Parquet stores the data, and Delta makes it a reliable table.

The Short says you can try this in a notebook in Databricks Free Edition. No paid features are needed.